Assignment: End-to-End Exploratory Data Analysis (EDA), Data Cleaning, Outlier Detection, and Data Preparation

Dataset: IBM HR Analytics Employee Attrition Dataset

Objective

You are hired as a Data Analyst in an HR department. The company wants to understand employee data quality, identify potential issues in the dataset, perform exploratory analysis, detect anomalies, and prepare the dataset for future Machine Learning models that predict employee attrition.

Your task is to perform a complete End-to-End EDA and Data Cleaning process and provide business insights.

Part 1: Understanding the Dataset

Task 1.1: Load the Dataset

Perform the following:

1. Load the dataset.
2. Display the first 10 records.
3. Display the last 10 records.
4. Find the number of rows and columns.
5. Display all column names.
6. Check the data type of each column.

Questions

1. How many records are present in the dataset?
2. How many features are available?
3. Which columns are numerical?

4. Which columns are categorical?

Task 1.2: Dataset Overview

Generate:

- Dataset summary
- Statistical summary
- Information summary

Questions

1. Which feature has the highest average value?
2. Which feature has the highest standard deviation?
3. Which feature has the minimum variability?

Part 2: Data Quality Assessment

Task 2.1: Missing Values Analysis

Calculate:

- Total missing values
- Percentage of missing values

Create:

- Missing value table
- Missing value bar chart

Questions

1. Which column contains the most missing values?
2. Which columns have no missing values?
3. How can missing values affect Machine Learning models?

Task 2.2: Duplicate Analysis

Perform:

1. Check duplicate records.
2. Count duplicates.
3. Remove duplicates.

Questions

1. How many duplicate records were found?
2. Why should duplicates be removed?

Task 2.3: Unique Value Analysis

For each column:

Calculate:

- Number of unique values
- Most frequent value

Questions

1. Which feature has the highest cardinality?
2. Which feature can be considered an identifier?

Part 3: Data Cleaning

Task 3.1: Missing Value Treatment

For numerical columns:

Apply:

- Mean Imputation
- Median Imputation

For categorical columns:

Apply:

- Mode Imputation
- Unknown Category Imputation

Questions

1. Which imputation method worked best?
2. Why is median preferred for skewed distributions?

Task 3.2: Data Type Validation

Check whether:

- Numerical columns are stored correctly.
- Date columns are in datetime format.
- Categorical columns are properly categorized.

Questions

1. Were any incorrect data types found?
2. How can incorrect data types impact analysis?

Part 4: Univariate Analysis

Task 4.1: Numerical Features Analysis

For each numerical feature:

Calculate:

- Mean
- Median
- Mode
- Variance
- Standard Deviation
- Minimum
- Maximum
- Range
- Quartiles

Create:

- Histogram
- KDE Plot
- Boxplot

Questions

1. Which feature is normally distributed?
2. Which feature is right-skewed?
3. Which feature is left-skewed?

Task 4.2: Categorical Features Analysis

For each categorical feature:

Calculate:

- Frequency Count
- Percentage Distribution

Create:

- Countplot
- Pie Chart

Questions

1. Which department has the most employees?
2. Which job role is most common?
3. Which category is least represented?

Part 5: Outlier Detection

Task 5.1: IQR Method

For every numerical column:

Calculate:

- Q1
- Q3
- IQR
- Lower Bound
- Upper Bound

Identify:

- Number of outliers

Create:

- Boxplots before treatment

Questions

1. Which feature has the most outliers?
2. Why are outliers important?

Task 5.2: Z-Score Method

Calculate:

Z-score for every numerical feature.

Identify observations where:

|Z-score| > 3

Questions

1. Compare IQR and Z-score methods.
2. Which method performs better on skewed data?

Part 6: Outlier Treatment

Apply all three methods:

Method 1

Remove Outliers

Method 2

Capping

Replace values outside bounds.

Method 3

Winsorization

Replace extreme values with percentile limits.

Questions

1. Which treatment preserved the most data?
2. Which treatment changed the mean significantly?
3. Which treatment would you recommend?

Part 7: Feature Engineering

Create at least five new features.

Examples:

Age Group

18–25 → Young

26–35 → Mid

36–45 → Senior

45+ → Veteran

Income Category

Low Income

Medium Income

High Income

Experience Category

Beginner

Intermediate

Experienced

Expert

Promotion Risk Score

Based on:

- YearsSinceLastPromotion
- JobLevel

Attrition Risk Score

Based on:

- Overtime
- DistanceFromHome
- MonthlyIncome

Questions

1. Which engineered feature seems most useful?
2. How can feature engineering improve model performance?

Part 8: Bivariate Analysis

Numerical vs Numerical

Create:

- Correlation Matrix
- Heatmap
- Scatterplots

Questions

1. Which features are highly correlated?
2. Which features show a strong positive relationship?
3. Which features show a negative relationship?

Numerical vs Categorical

Create:

- Grouped Statistics
- Boxplots
- Violin Plots

Questions

1. Which department has the highest salary?
2. Which job role shows the highest income variation?

Categorical vs Categorical

Create:

- Crosstab
- Heatmap

Questions

1. Is attrition associated with overtime?
2. Is attrition associated with business travel?

Part 9: Data Transformation

Apply:

StandardScaler

MinMaxScaler

RobustScaler

Compare:

- Mean
- Standard Deviation
- Distribution

Questions

1. Which scaler handled outliers best?
2. Which scaler is suitable for neural networks?

Part 10: Skewness Analysis

Calculate skewness for all numerical features.

Apply:

- Log Transformation
- Square Root Transformation

Compare skewness before and after.

Questions

1. Which transformation reduced skewness the most?
2. Why is skewness problematic?

Part 11: Data Engineering Quality Checks

Create validation checks for:

Null Values

Example:

MonthlyIncome should not be null.

Negative Values

Example:

Age < 0

YearsAtCompany < 0

Business Rules

Examples:

YearsAtCompany ≤ TotalWorkingYears

MonthlyIncome > 0

Age > 18

EmployeeNumber should be unique

Data Type Rules

Age → Integer

MonthlyIncome → Numeric

Department → String

Questions

3. Which business rules failed?
4. How would you automate these checks in production?

Part 12: Machine Learning Readiness Assessment

Create a report containing:

Columns to Drop

Explain why.

Columns to Encode

Choose:

- Label Encoding
- One Hot Encoding

Columns to Scale

Explain why scaling is required.

Columns with High Correlation

Recommend whether to keep or remove them.

Final Feature List

Prepare the dataset for machine learning.

Part 13: Business Insights

Generate at least 10 business insights.

Examples:

1. Employees with overtime show higher attrition.
2. Employees with low income leave more frequently.
3. Long travel distance increases attrition risk.
4. Certain job roles have higher attrition rates.
5. Employees receiving fewer promotions are more likely to leave.

Support every insight with a visualization.

Bonus Challenge (20 Marks)

Create a reusable Python class named:

```
class HRDataAnalyzer:
    pass
```

The class should automatically:

- Generate Summary Statistics
- Detect Missing Values
- Detect Duplicates
- Detect Outliers
- Create Visualizations
- Generate Data Quality Report
- Generate Correlation Report
- Generate Machine Learning Readiness Report

Submission Requirements

1. Jupyter Notebook (.ipynb)
2. Cleaned Dataset (.csv)
3. Data Quality Report (.pdf)
4. Machine Learning Readiness Report (.pdf)
5. Business Insights Report (.pdf)
6. Presentation (8–10 Slides)